

$$\begin{aligned}
 &= ABC + AB'C + ABC + ABC + ABC + ABC \text{ (Theorem 4(a))} \\
 &= ABC + A'BC + AB'C + AB'C + ABC + ABC \text{ (Postulate 3(a))} \\
 &= (ABC + A'BC) + (AB'C + AB'C + ABC + ABC) \text{ (Theorem 4(a))} \\
 &= (\cancel{ABC} + A'BC) + ((AB'C + AB'C) + (ABC + ABC)) \text{ (Theorem 4(a))} \\
 &= (A \cdot BC + A' \cdot BC) + ((AB'C + AB'C) + (ABC + ABC)) \text{ (Theorem 4(b))} \\
 &= (BC \cdot A + BC \cdot A') + ((AB'C + AB'C) + (ABC + ABC)) \text{ (Postulate 3(b))} \\
 &= (BC \cdot (A + A')) + (AB' \cdot (C + C) + AB \cdot (C + C)) \text{ (Postulate 4(a))} \\
 &= BC \cdot (A + A') + (AB' \cdot (C + C) + AB \cdot (C + C)) \text{ (Postulate 3(a))} \\
 &= BC \cdot 1 + (AB' \cdot 1 + AB \cdot 1) \text{ (Postulate 5(a))} \\
 &= BC + (AB' + AB) \text{ (Postulate 2(b))} \\
 &= BC + (AB + AB') \text{ (Postulate 3(a))} \\
 &= BC + (A \cdot (B + B')) \text{ (Postulate 4(a))} \\
 &= BC + A \cdot 1 \text{ (Postulate 5(a))} \\
 &= BC + A \text{ (Postulate 2(b))}
 \end{aligned}$$

$= BC + A \cdot 1$ (Postulate 5(a))
 $= BC + A$ (Postulate 2(b))

2-13: Express the following functions in a sum of minterms and a product of maxterms.

(1) $F(A, B, C, D) = D(A+B) + B'D$

[illegible]